

Acceleration Control for a Differential-Thrust Quad-Plane Interceptor

Study Presentation

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Project Subject: Acceleration Control for a Differential-Thrust Quad-Plane Interceptor



Problem Statement

Objective:

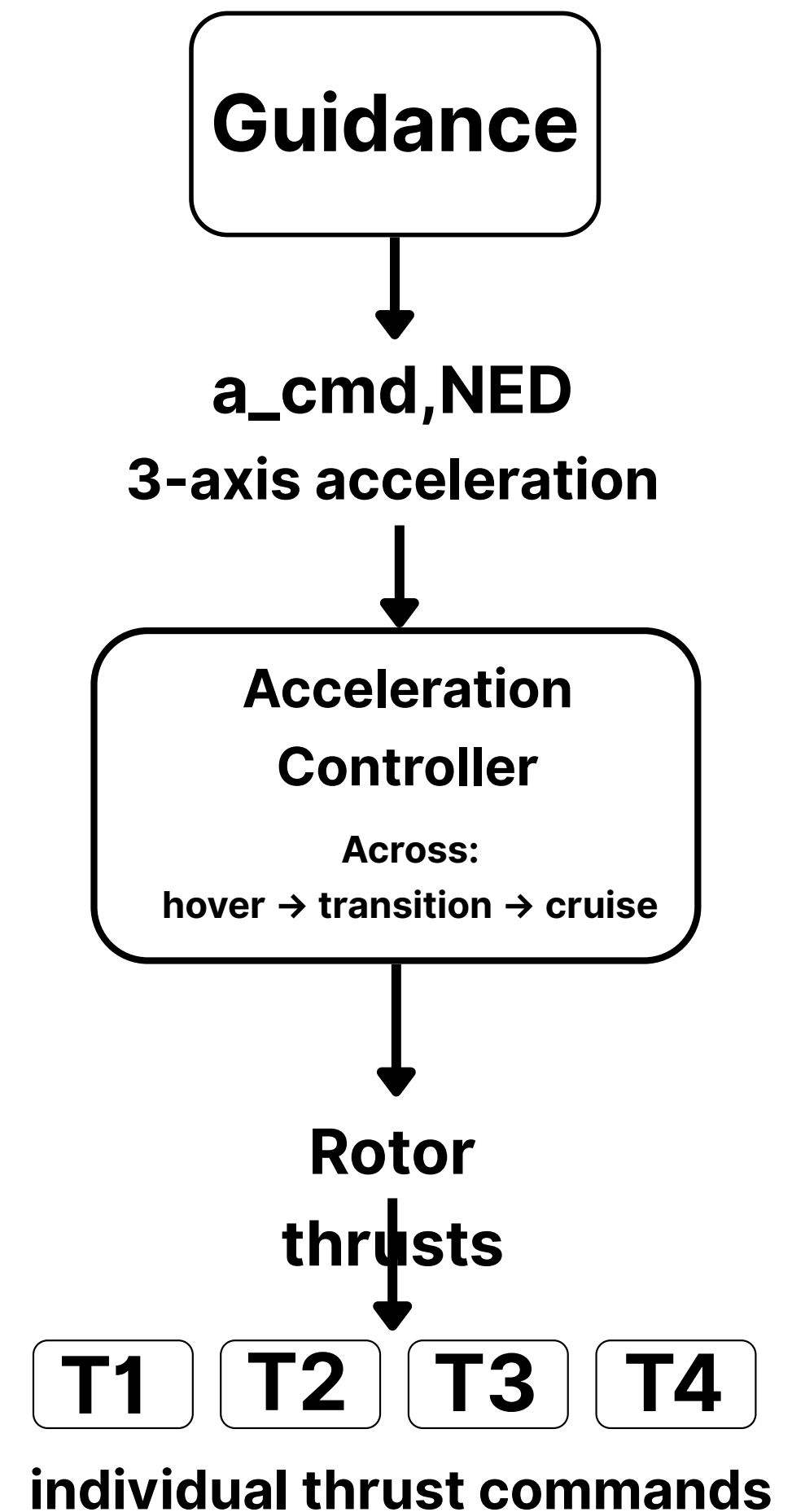
Design a regime-spanning acceleration controller for a quad-plane interceptor.

Platform challenge:

- Fixed wing + four fixed rotors
- No elevator, aileron, rudder, or tilt mechanism
- Roll, pitch, and yaw generated only by differential thrust
- Guidance input: 3-axis inertial acceleration command
- Controller output: individual rotor thrust commands

Core challenge:

The same four rotors generate both force and moments.



Platform Physics - Hover vs Cruise

Hover regime

Force generation:

- Vertical acceleration from collective rotor thrust
- Lateral acceleration by tilting the thrust vector

Moment generation:

- Roll / pitch / yaw from differential thrust

Approximation:

$$F_z = \sum T_i$$

$$a_z \approx \sum T_i / m - g$$

Key change:

- In hover, rotors directly create vertical force.
- In cruise, rotors mainly create attitude moments that redirect wing lift.

Cruise regime

Force generation:

- Wing provides most of the lift
- Vertical / lateral acceleration from pitch & roll attitude
- Total thrust mainly balances drag / speed

Moment generation:

- Still only differential thrust
- No elevator / aileron / rudder

Approximation:

$$L \approx 0.5 \cdot \rho \cdot V^2 \cdot S \cdot C_L(\alpha)$$

Static Pitch Stability

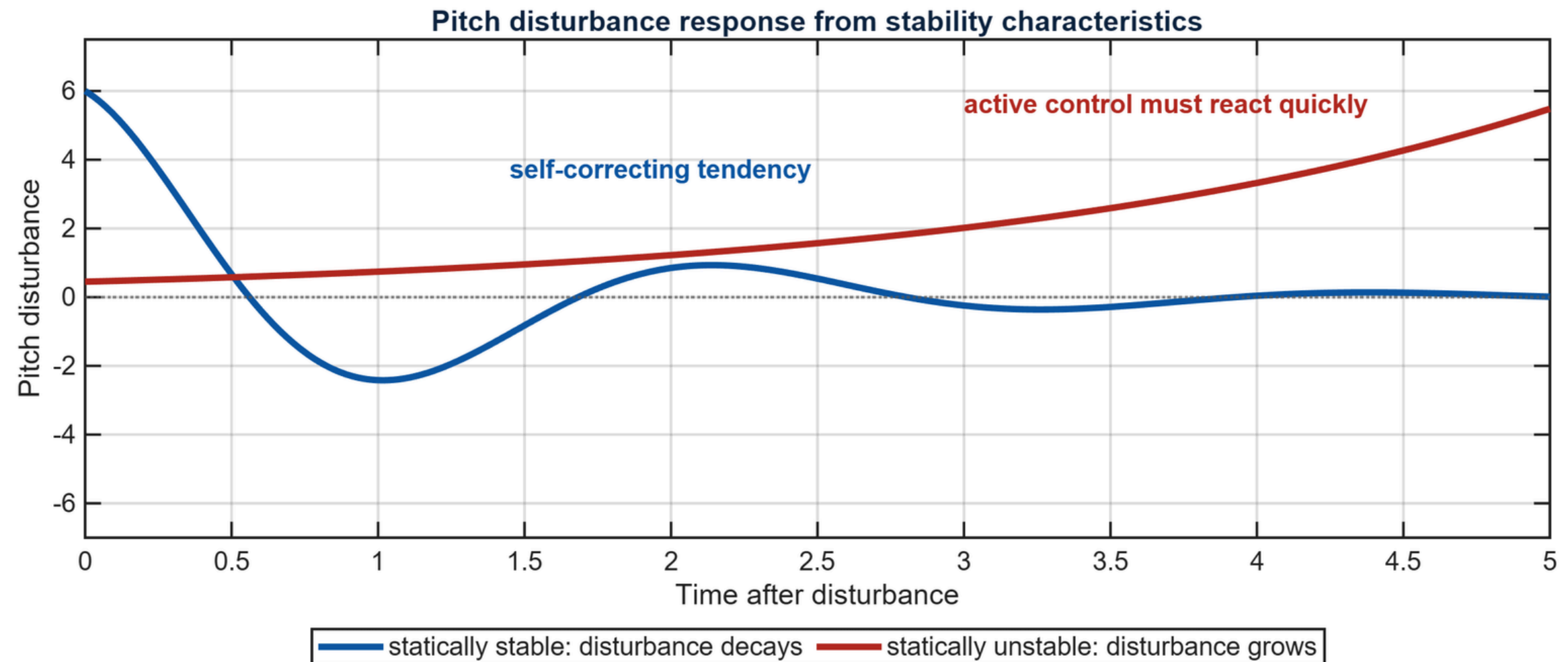
Question:

- Is the aircraft statically stable in pitch during fixed-wing flight?

If it is unstable:

- disturbance grows without fast active control
- controller delay becomes critical
- failsafe and update rate are safety-critical

Design implication: unstable pitch demands high update rate, low latency, and strong failsafe systems.



Differential-Thrust Authority vs Airspeed

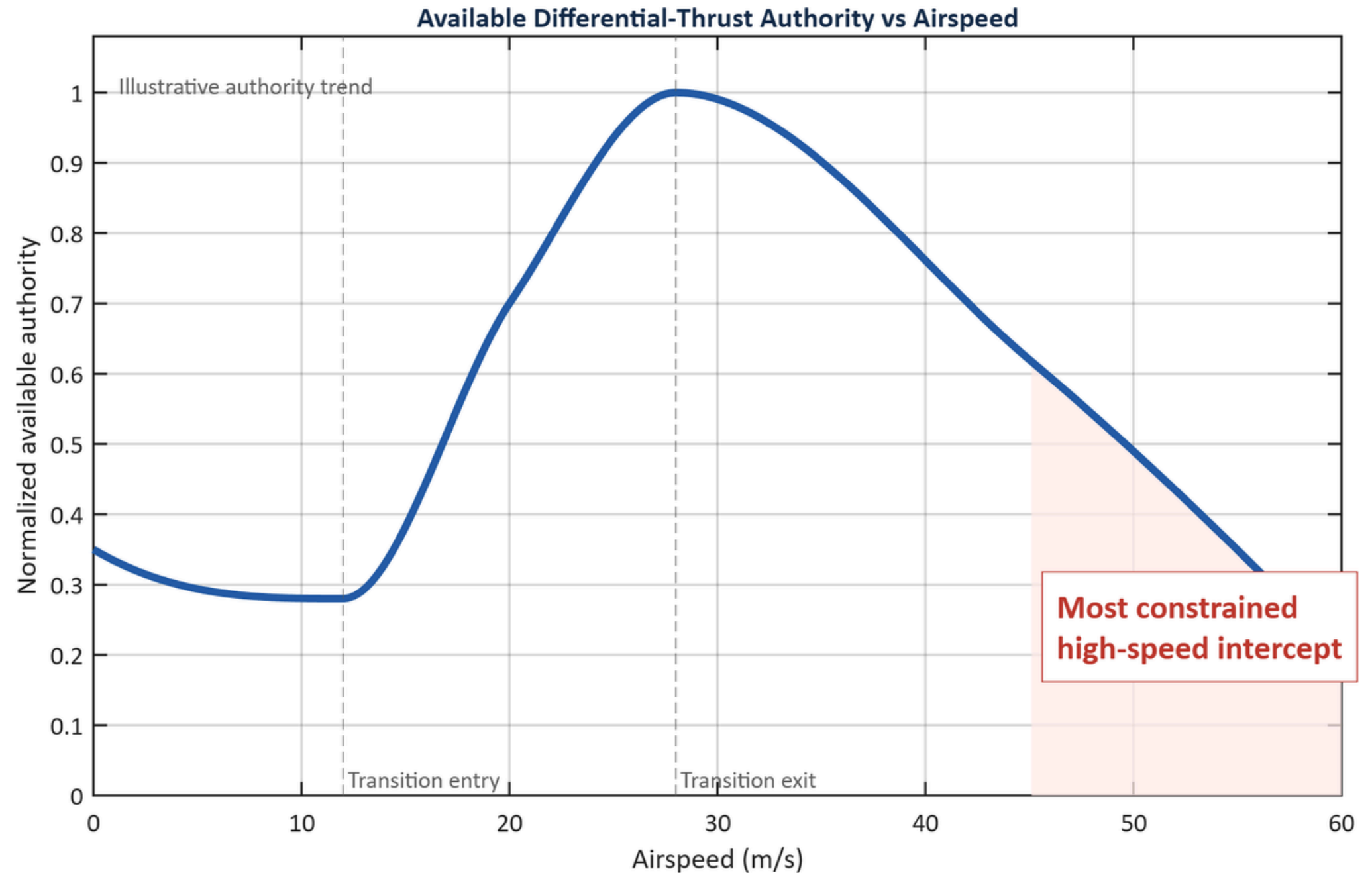
Differential thrust margin

$$T_i = T_{\text{trim},i} \pm \Delta T_i$$

$$0 \leq T_i \leq T_{\text{max}}$$

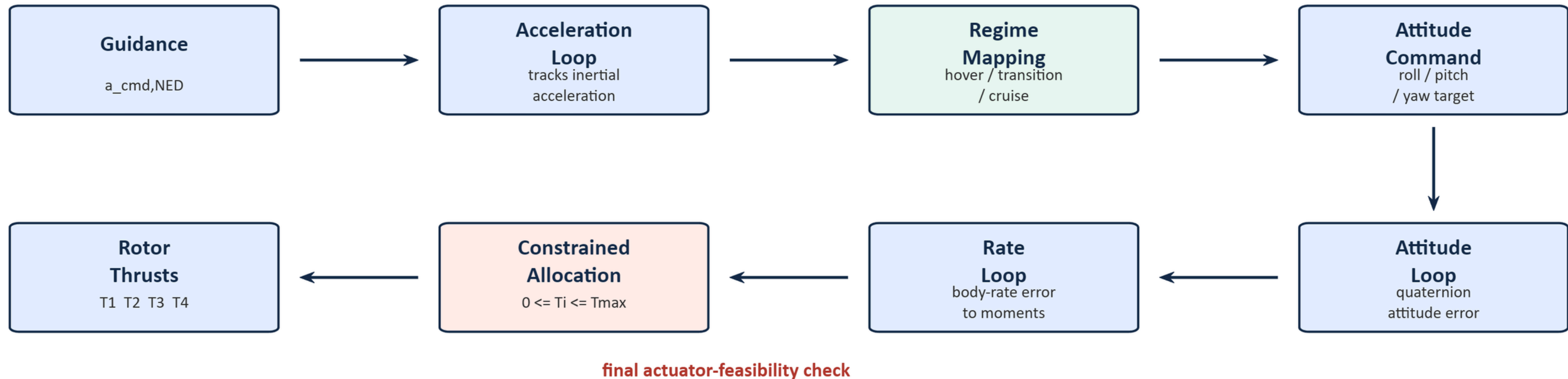
Authority depends on:

- rotor thrust margin
- airspeed / drag demand
- wing lift contribution
- actuator bandwidth



Limited moment authority → limited attitude response → limited acceleration tracking

Proposed Control Architecture



Design choices:

- cascaded loops for bandwidth separation
- smooth regime-dependent mapping
- quaternion attitude loop
- constrained allocation

Quaternion attitude representation avoids the gimbal-lock singularity near $\pm 90^\circ$ pitch — relevant here because aggressive intercept maneuvers can push pitch attitude into that range."

INDI (Incremental Nonlinear Dynamic Inversion)

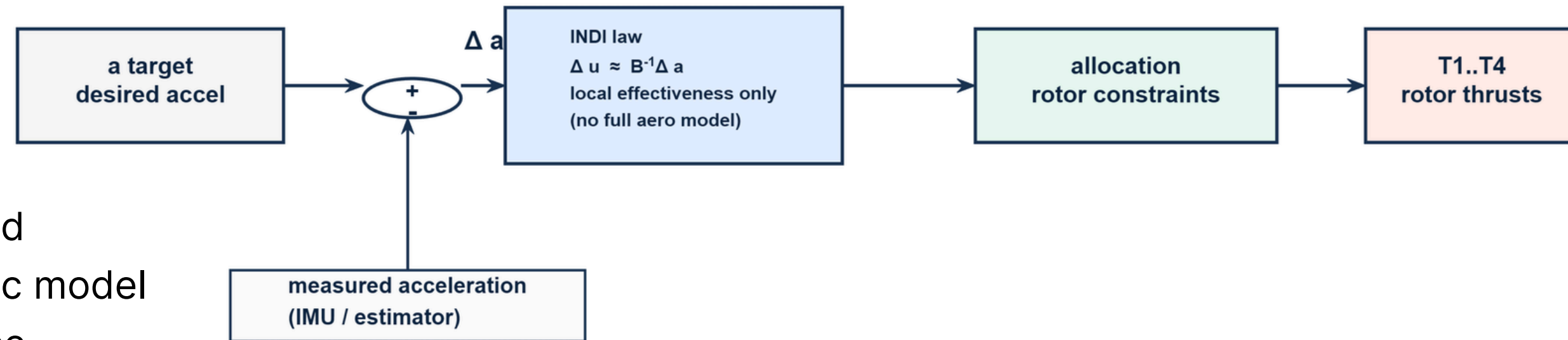
INDI acceleration loop: use measured response to command only the increment

Core idea:

use target acceleration minus measured acceleration

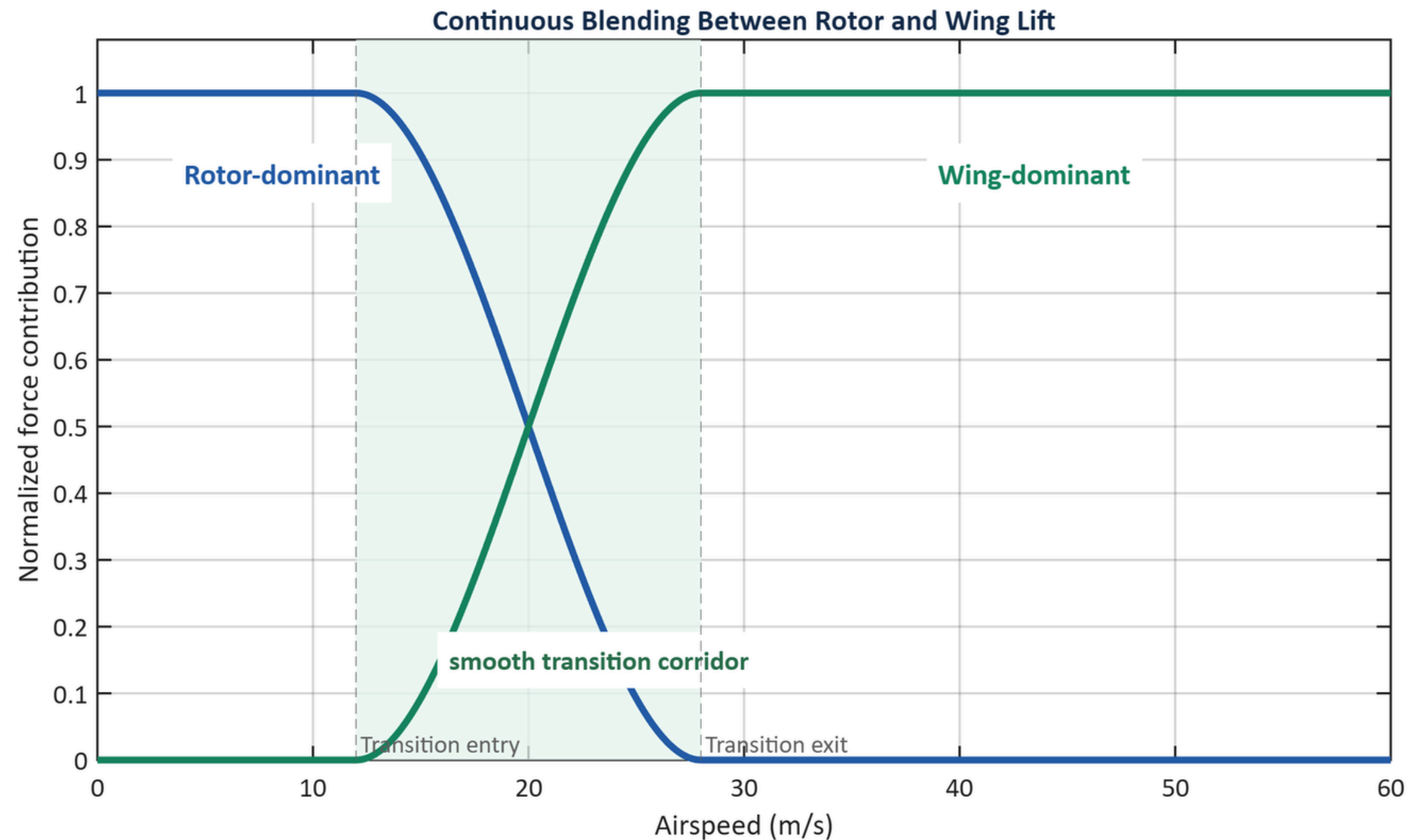
Why INDI fits:

- calculates only the incremental command
- reduces dependence on full aerodynamic model
- useful across hover, transition, and cruise



Key idea: update the command from measured acceleration error, not from a perfect full aerodynamic model

Transition Handling: Continuous Blending



Key idea: estimate wing lift and command rotors only for the remaining required force.

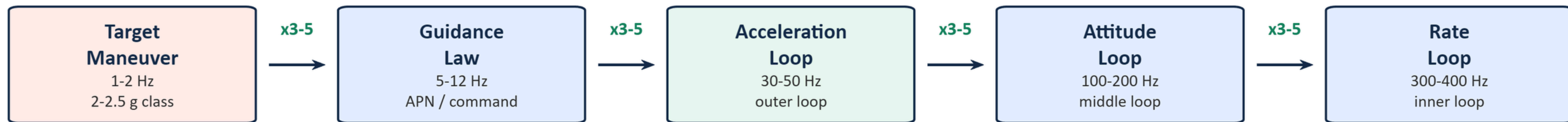
Why smooth blending?

- avoids hard mode-switching
- shares lift between rotors and wing
- prevents double-counting of vertical force

Discrete switching (risk: command discontinuity at low authority),
Gain scheduling (similar, less explicit)
Continuous blending (chosen: physically motivated, smooth)."

Bandwidth Chain

Candidate loop-rate targets for intercept-quality acceleration tracking.



Why it matters:

- Attitude bandwidth limits acceleration bandwidth in cruise
- Motor / ESC dynamics bound the inner rate loop
- Margins must be checked against actuator authority

Deep Dive: Control Allocation Under Saturation

Nominal mixing relation

$$u = B T$$

$$u = [T_{\text{collective}}, \tau_{\text{roll}}, \tau_{\text{pitch}}, \tau_{\text{yaw}}]^T$$

$$T = [T_1, T_2, T_3, T_4]^T$$

$$B = \begin{bmatrix} 1 & 1 & 1 & 1 \\ y1 & y2 & y3 & y4 \\ -x1 & -x2 & -x3 & -x4 \\ k s1 & k s2 & k s3 & k s4 \end{bmatrix}$$

xi, yi: rotor position | si: spin direction sign | k: reaction-torque coefficient

Physical rotor limits

$$0 \leq T_i \leq T_{\text{max}}$$

If nominal mixing gives:

$$T_i < 0 \quad \text{or} \quad T_i > T_{\text{max}}$$

**then u_{cmd} is infeasible.
The allocator must choose
the best achievable command.**

Constrained reallocation (not naive clipping)

$$T^* = \arg \min_T \|W_u (B T - u_{\text{cmd}})\|_2^2 + \rho \|T - T_{\text{trim}}\|_2^2$$

Key point:

- Nominal mixing is valid only if all rotor commands remain feasible.
- If T_i violates the limits, the requested force/moment vector is infeasible.

$$\text{subject to } 0 \leq T_i \leq T_{\text{max}}$$

Saturation Priority: What Should Degrade First?

Priority under saturation

1. Preserve roll / pitch moment authority

→ attitude and acceleration tracking

2. Preserve minimum collective thrust

→ flight safety / altitude support

3. Preserve yaw moment as much as possible

→ heading control

4. Relax excess forward thrust / speed demand first

→ acceptable short-term tradeoff during intercept

Regime-aware priority:

- Hover: collective thrust is safety-critical
- Cruise: moment authority is critical

For interception;

losing some speed is recoverable;

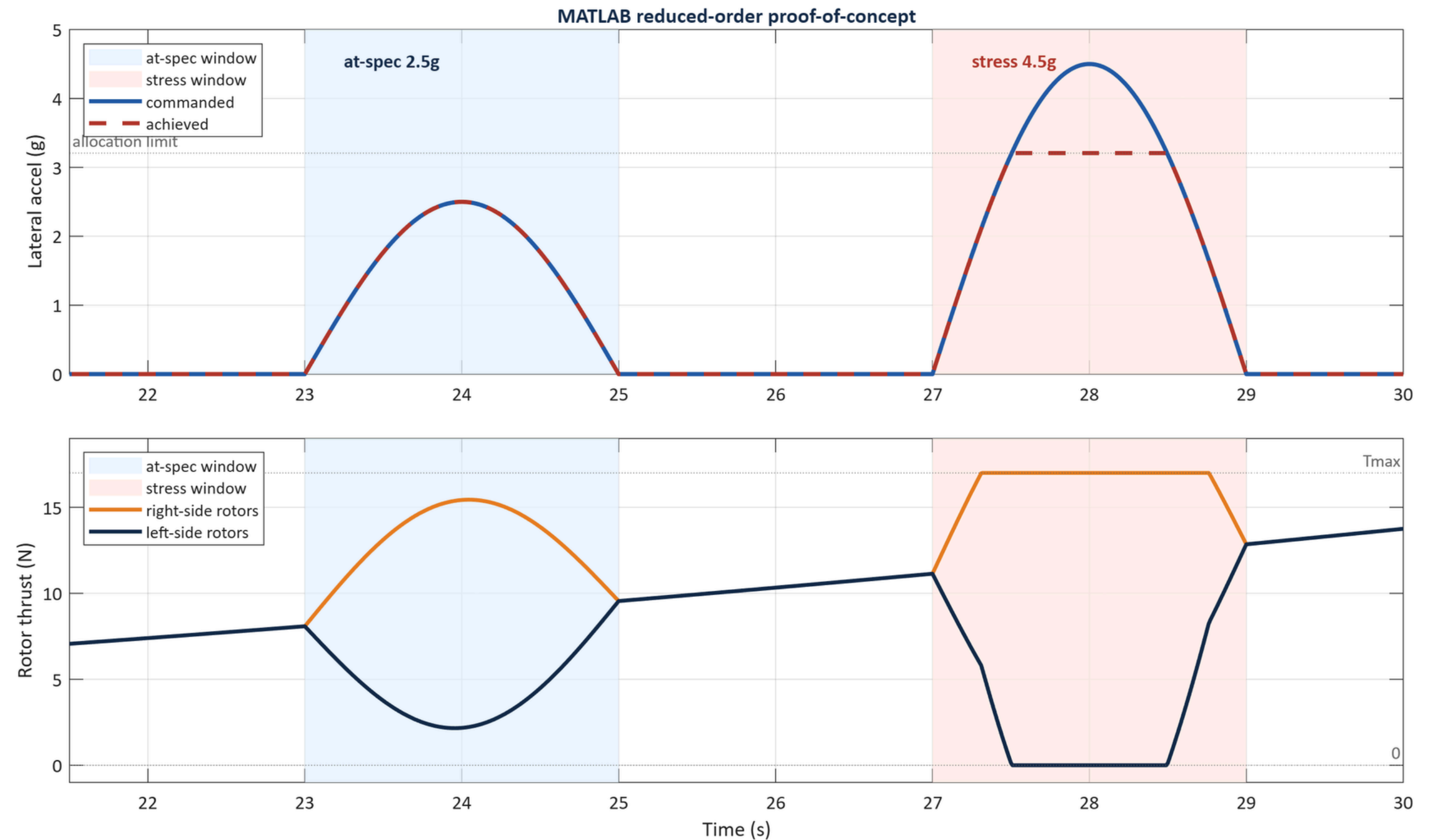
losing attitude authority can mean losing the target.

MATLAB Proof-of-Concept: Constrained Allocation

Scope:
Reduced-order lateral-axis allocation demo

Key results:

- 2.5g command is tracked
- 4.5g stress command is gracefully limited
- Rotor thrusts remain inside physical bounds



Transition Corridor & Avoiding Double-Counting

Transition corridor:

- blending variable based on airspeed
- hover: rotor-dominant vertical force
- cruise: wing-dominant lift generation

Rotor vertical command = required vertical force - estimated wing lift

$$F_{\text{rotor},z} = F_{\text{required},z} - \lambda(V) L_{\text{wing,estimated}}$$

Avoiding double-counting:

- estimate wing lift contribution
- command rotors only for the remaining required force
- keep force command continuous through transition

$\lambda(V)$: 0 in hover, 1 in cruise

Interception-Specific Reality Check

Interception reality: acceleration is authority-limited



Commanded acceleration must be checked against actuator feasibility

Interception constraints:

- no elevator / aileron / rudder
- all moments from differential thrust
- high-speed margin is limited

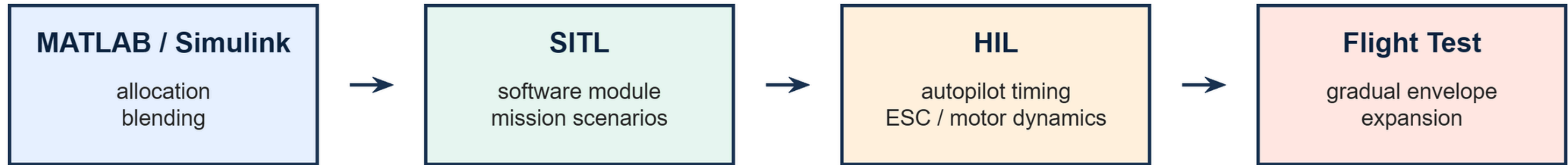
Main takeaway:

Acceleration tracking is limited by available differential-thrust margin.

Design implication:

Acceleration commands must be feasibility-aware.

Implementation Path



Main tuning focus: transition blending, saturation priority, inner-loop bandwidth

Step 1 - MATLAB / Simulink

- validate allocation logic
- tune blending and saturation priorities
- test hover, transition, cruise cases

Step 2 - SITL

- implement controller as a software module
- connect guidance acceleration command
- test mission-level intercept scenarios

Step 3 - HIL

- include real autopilot, ESC, motor dynamics
- check timing, delays, and actuator limits

Step 4 - Flight Test

- start with hover and low-speed transition
- expand envelope gradually
- validate acceleration tracking and saturation behavior



Summary

Objective coverage:

1. Regime-spanning acceleration control

→ cascaded acceleration, attitude, and rate loops

2. Unified rotor thrust allocation

→ smooth regime mapping and constrained allocation

3. Saturation handling

→ priority-based degradation under rotor limits

4. Acceleration bandwidth

→ bandwidth separation from guidance to rate loop

Main contribution:

A feasibility-aware acceleration control architecture for a differential-thrust quad-plane interceptor.





References

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 - [4] Johansen and Fossen, control allocation survey, 2013.
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